The Art of Probability

FOR SCIENTISTS AND ENGINEERS

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7

Maximum Entropy

"There is no doubt that in the years to come the study of entropy will become a permanent part of probability theory;. . ."

A. I. Khinchin (in 1933) [K,p.2]

7.1 What is Entropy?

The main purpose for studying entropy is that it provides a new tool for assigning initial probability distributions. The method is consistant with our symmetry approach but also includes nonuniform distributions. Thus it is a significant extension of the principle of symmetry.

Just as the word "probability" is used in many different senses, the word "entropy" also has several, perhaps incompatible, meanings. We therefore need to examine these different usages closely since they are the source of a great deal of confusion-especially as most people have only a slight idea of what "entropy" means. We will examine four different meanings that are the more common usages of the word. Since this is not a book on thermodynamics or statistical mechanics the treatment must, of necessity, be fairly brief and superficial in the thermodynamically related cases.

The word entropy apparently arose first in classical thermodynamics. Classical thermodynamics treats state variables that pertain to the whole system, such as pressure, volume and temperature of a gas. Besides the directly measurable variables there are a number of internal states that must be inferred, such as *entropy* and *enthalpy*. These arise from the imposition of the law of the conservation of energy and describe the internal state of the system. A mathematical equation that arises *constantly is*

$$dH = dQ/T$$

where ***dH*** is the change (in the sense of engineers and physicists) in entropy, ***T*** refers to the absolute temperature, and ***dQ*** is the quantity of heat transferred at temperature ***T.*** From the assumed second law of thermodynamics (that heat does not flow spontaneously from cold to hot) it is asserted that this quantity can only increase or stay the same through any changes in a closed system. Again, classical thermodynamics refers to global properties of the system and makes no assumptions of the detailed micro structure of the material involve.

Classical statistical ***mechanics*** tries to model the detailed structure of the materials, and from this model to predict the rules of classical thermodynamics. This is "micro statistics" in place of "macro statistics" of classical thermodynamics. Crudely speaking, matter is assumed to be little hard, perfectly elastic, balls in constant motion, and this motion of the molecules bouncing against the walls of the container produces the pressure of the gas, for example. Even a small amount of gas will have an enormous number N of particles, and we imagine a phase space whose coordinates are the position and velocity of each particle. This phase space is a subregion of a 6N dimensional space, and in the absence of any possibility of making the requisite measurements, it is ***assumed*** that for a fixed energy every small region in the phase space has the same probability as any other (compare the Poincare theory of small causes, large effects). Then average values over the phase space are computed. Boltzmann broke up the phase space into a large number of very small regions and found that the quantity

$$H = k \log \frac{1}{P}$$

seemed to play a role similar to the classical concept of entropy (k is Boltzmann's constant and P is the probability associated with any one of the equally likely small regions in the phase space with the given energy.

Boltzmann attempted to prove the famous Boltzmann's "H theorem" which states that with increasing time H could not decrease but would generally increase. It was immediately pointed out by Loschmidt that since Boltzmann began with Newtonian mechanics, in which all the equations of motion that he used are reversible, then it is impossible from this ***alone*** to deduce irreversibility. A lot of words have been said about the probabilities of the states when all the velocities of the particles are reversed (to make things go backwards in time), and it is usually claimed that these states are much less likely to occur than those in the original direction-though so far as the author knows there never has been any careful mathematical proof. only the argument that this must be so in order for the theorem they wanted to be true! Kac gave an interesting simple counter example to the proof using only a finite system that was clearly periodic and reversible. (See Wa, pp.396-404)

This, then, is the second entropy that is widely used, and is thought to be connected with the "mixed-up-ness" of the system.

Gibbs, in trying to deal with systems which did not have a fixed energy introduced the "grand cannonicai ensemble" which is really an ensemble of the phase spaces of different energies of Boltzmann. In this situation he deduced an entropy having the form

$$H = \sum_i p(i) ln \frac{1}{p(i)}$$

where the p(i) are the probabiiites of the phase space ensembles. This, then, is still a third kind of physical entropy.

Shannon while creating information theory found an expression of exactly the same form, and called it entropy, apparently on the advice of von Neumann, ("You should call it 'entropy' for two reasons: first, the function is already in use in thermodynamics under that name; second, and more importantly, most people don't know what entropy really is, and if you use the word 'entropy' in an argument you will win every time!") [L-T, p.2]

But the identity of mathematical form does not imply identity of meaning. For example, Galileo found that failing bodies obey the formula

$$s = \frac{g}{2} t^2$$

and Einstein found the famous equation

$$E = mc^2$$

but few people believe that the two equations have much in common except that both are parabolas. As the derivation of Shannon's entropy in the next section will show, the variables in his equation bear little resemblence to those of Gibbs or Boltzmann. However, from Shannon's entropy Tribus has derived all the basic laws of thermodynamics [L-T] (and has been ignored by most thermodynamicists).

While the identity of form cannot prove the identity of meaning of the symbois, it does mean that the same formal mathematics can be used in both areas. Hence we can use many of the formal mathematical results and methods, but we must be careful in the interpretations of the results.

7.2 Shannon's Entropy

Let X be a discrete random variable with probabilities P(i), (i = 1.2,. . . , n). We shall show in a few moments that when dealing with "information" the standard assignment for the amount of information, $I\{p(i)\}$, to be associated with an outcome of probability p(i), is measured by the "surprise," the likelihood of the event, or the information gained, which is

$$I\{p(i)\} = \log \frac{1}{p(i)} = -\log p(i)$$

In the certain event P = 1, and no new information is obtained when it happens.

The result of this assignment is that the ***expected*** value of the information f(X) for a random variable X is given by the formula,

$$H(X) = E\{I(X)\} = \sum_{i=1}^{n} p(i) \log \frac{1}{p(i)} \qquad (7.2\text{-}1)$$

In ***information theory*** this is called the entropy. It is usually labeled H(X), or sometimes H(P).

Information theory usually uses the base 2 for the logarithm system. which we do in this chapter unless the contrary is explicitly stated. Logs co the base e will be denoted, as usual, by ln.

Kullback [Ku] has written a book (1959) with the title, ***Information Theory and Statistics,*** which shows many of the connections between the two fields.

This assignment of values to a random variable X is one of a number of different ***intrinsic assignments of*** values to the outcomes of a trial of a random variable, "intrinsic" meaning that the values come direct/y from the probabilities and have no ***extrinsic*** source. The general case of intrinsic assignments is the assignment of the value of the random variable as some function /(p(i)). In contrast, dealing with the time to first failure for a run of n trials gives an ***extrinsic*** but natural assignment of the index n as the values for the possible outcomes of probabilities qp^{n-1}, $(n = 1,2,. . .)$.

The entropy can also be viewed in a different way; since

$$\text{P(i)} \log \frac{1}{p(i)} = \log \left\{ \begin{matrix} 1 \\ P\ O \end{matrix} \right\}^{p(i)}$$

we have

$$H(X) = -\log \prod_{i=1}^{n} p(i)^{p(i)}$$

and taking antilogs of both sides we have

$$2^{-H(X)} = \prod_{i=1}^{n} p(i)^{p(i)} \tag{7.2-2}$$

which resembles a weighted geometric mean of the probabilities. Because of this logarithmic structure Shannon's entropy has many of the properties of both the arithmetic and geometric means.

Example 7.2-1 An Entropy Computation

Given a random variable X with the probability distribution $p(1) = 1/2$, $p(2) = 1/4$, $p(3) = 1/8$, and $p(4) = 1/8$, the entropy of the distribution is

$$H(X) = (1/2)\log 2 + (1/4)\log 4 + (1/8)\log 8 + (1/8)\log 8$$

$$= (1/2)(1) + (1/4)(2) + (1/8)(3) + (1/8)(3)$$

$$= 1/2 + 1/2 + 3/8 + 3/8 = 7/4$$

Rom any finite probability distribution we obtain the entropy as a single number-much as the mean is a single number derived from a distribution.

The entropy function measures the amount of information contained in the distribution, and is sometimes called the negentropy since it is thought to be the negative of the usual entropy which in turn is supposed to measure the "mixed-up-ness" of a distribution. We have already exposed the weakness of the analogy of Shannon's entropy with the entropies of physics, but see [T],[L-T]. For biology see [Ga].

Since we will be using Shannon's entropy extensively we need to make some reasonable derivation of his entropy function. Let there be two independent random variables X and Y with outcomes x_i and y_j (say the roll of a die and the toss of a coin). Bow much information is contained in the observed outcome $x_i y_j$? If $I(.)$ is the measure of the amount of information then we believe that for independent random variables the total information is the sum of the information contained in the individual outcomes, that is

$$I(x_i y_j) = I(x_i) + I(y_j) \tag{7.2-3}$$

Think this over carefully before accepting it as fitting your views of information.

Equation (7.2-3) is the standard Cauchy functional equation when we also suppose that the measure of information $I(p)$ is a non-negative continuous function of its argument.

To study the solution of Cauchy's equation we drop the subscripts. Consider the functional equation (defines a function rather than a number)

$$f(xy) = f(x) + f(y) \tag{7.2-4}$$

where $f(x)$ is a continuous nonnegative function of x. Suppose that y = x, then we have from equation (7.2-4)

$$f(x^2) = 2f(x)$$

Next, let y = x^2. Then from equation (7.24) we get

$$f(x^3) = 3f(x)$$

and in general, by induction, we have for ail integers $n > 0$,

$$f(x^n) = nf(x)$$

Now if we write $x^n = z$ we have

$$f(z) = nf(z^{1/n})$$

or rewriting this

$$f(z^{1/n}) = (1/n)f(z)$$

Proceeding as before, we can show that for all rational numbers p/q

$$f(z^{p/q}) = (p/q)f(z)$$

By the continuity assumption on the solution of the equation since the equality holds for all rational numbers it holds for ail real numbers, $x > 0$ and y

$$f(x^y) = yf(x) \tag{7.2-5}$$

The relationship (7.2-5) suggests that the solution is $f(x) = \log x$ to some base. To prove that this solution is unique to within a constant multiplier (effectively the base chosen for the logs) we assume that there is a second solution g(x), and then consider the difference

$$f(x) - kg(x) = \log_2 x - kg(x)$$

where, of course from (7.2-5) $g(x^y) = yg(x)$. We have for the difference

$$\log x - \mathrm{kg(t)} = (1/y)\{\log x^y - kg(x^y)\}$$

We fix in our minds some $x = x_0$, (not 0 or 1), and choose a particular k so that at $x = x_0$ the left hand side is zero (so that the scaled second solution has a common point with the log solution at $x = x_0$). Hence we set

$$k = \frac{\log x_0}{g(x_0)}$$

We have on the right

$$\log x_0^y - kg(x_0^y) = 0$$

Now for any z other than 0 and 1 there is a y such that

$$z = x_0^y \qquad \text{namely} \qquad y = \frac{\log z}{\log x_0}$$

and we have

$$\log z = kg(z) = g(z)\frac{\log_b x_0}{g(x_0)}$$

Therefore any other solution g(r) is proportional the $\log z$; and the solutions differ only with respect to the logarithm base used.

It is conventional to use, in information theory, the base 2 so that a simple binary choice has unit information; it is only a difference in the units used to measure information that is being determined by the choice of base 2. The standard equation

$$\log x = \frac{\log_b x}{\log_b 2} = (\log_2 b)(\log_b x)$$

means that every log system is proportional to every other, and the conversion from one to another is simple.

Since by assumption $I(p) \geq 0$ it is necessary to use a negative value for k so we finally choose our solution of equation (7.2-3) as

$$f(p) = \log \frac{1}{p} = -\log p \tag{7.2-6}$$

Now that we have the measure of information for any outcome of probability p, we examine it to see its reasonableness. The less likely the event is to happen the more the information, the more the surprise, the greater the reduction in ***uncertainty.*** The certain event $p = 1$ has no information, while the impossible event has infinite information, but it never occurs! See below, equation (7.2-g).

The average information contained in a distribution $p(i)$ is clearly the weighted average of log{l/p(i)} (the expected value). From (7.2-6) we have

$$H(X) = \sum_{i=1}^{n} p(i) \log \frac{1}{p(i)} = -\sum_{i=1}^{n} p(i) \log p(i) \tag{7.2-7}$$

which is the ***entropy*** of the distribution. The entropy is measured in units of bits per trial.

When we have a binary choice then we often write the argument as *p (or q)*

$$H(p) = p \log \frac{1}{p} + q \log \frac{1}{q} = H(q)$$

as as matter of convenience since the entropy then depends on the single number p (or q).

We need to examine the shape (graph) of the typical term of the entropy function as a function of the probability p. There is clearly trouble at p = 0. To evaluate this indeterminate form we first shift to logs to the base e and later rescaie.

$$\lim_{p \to 0} p \ln \frac{1}{p} = \lim_{p \to 0} \left[\frac{\ln(1/p)}{1/p} \right]$$

We can now apply l'Hopital's rule,

$$\lim_{p \to 0} \left[\frac{-1/p}{-1/p^2} \right] = \lim_{p \to 0} p = 0 \qquad (7.2\text{–}8)$$

and we see that at $p = 0$ the term $p \ln(1/p) = 0$. Thus while the impossible event has infinite information the limit as p approaches 0 of the probability of its occurrence multiplied by the amount of information is still 0. We also see that the entropy of a distribution with one $p(i) = 1$ is 0; the outcome is certain and there is no information gained.

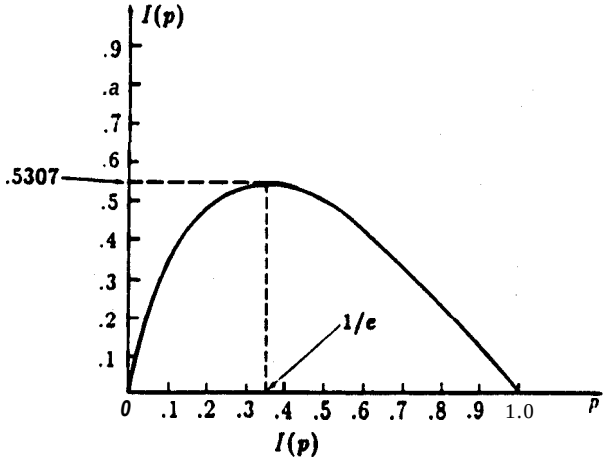


FIGURE 7.2-1

For the slope of the curve y = $p \ln(1/p)$ we have

$$y' = -1 - \ln p$$

and it is infinite at $p = 0$. The maximum of the curve occurs at p = l/e. Thus, rescaling to get to logs base 2, we have the Figure 7.2-1.

7.3

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Exercises 7.2

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7.2-3 Com
7.2-4 Show
7.2-5 For
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7.2-6 Let
$p_n = (1/n)/$
distribution
7.2-7 Usin

In coding theory it can be shown that the entropy of a distribution provides the lower bound on the average length of any possible encoding of a uniquely decodable code for which one symbol goes into one symbol. It also arises in an important binomial coefficient inequality

$$\sum_{k=0}^{an} C(n,k) \le 2^{nH(a)} \tag{7.2-9}$$

where $a < 1/2$ and provided n is made sufficiently large. Thus the entropy function plays an important role in both information theory and coding theory.

Example 7.2-2 The Entropy of Uniform Distributions

How much information (on the average) is obtained from the roll of a die? The entropy is

$$H(X) = \sum_{i=1}^{6} \frac{1}{6} \log 6$$

$$= \log 6 = 2.584962 \ldots \text{ bits/roll}$$

Similarly, for any uniform choice from n things the entropy is

$$H(X) = \log n \text{ bits/trial}$$

which gives a reasonable answer. In the cases where $n = 2^k$ you get exactly k bits of information per trial.

Exercises 7.2

7.2-1 Show that (7.2-9) holds at $a = 1/2$.

7.2-2 Compute the entropy of the distribution p(i) = 1/8, (i = 1, 2, . . . ,8).

7.2-3 Compute the entropy of $p(i) = 1/3, 1/3, 1/9, 1/9, 1/9$.

7.2-4 Show that $\int_0^1 I(p)\, dp = 0.3606\ldots$

7.2-5 For the probabilities 1/3, 1/4, 1/6, 1/9, 1/12, 1/18 show that the entropy is $8/9 + (11/12) \log 3 \sim 2.34177$.

7.2-6 Let $S_N = \sum_{n=1}^{N} 1/n$. Then Zipf's distribution has the probabilities $p_n = (1/n)/S_N$. If $T_N = \sum_{n=1}^{N} (\log n)/n$ show that the entropy of Zipf's distribution is $H(N) = T_N/S_N + \log S_N$.

7.2-7 Using Appendix A.1 estimate the entropy of Zipf's distribution.

7.3 Some Mathematical Properties of the Entropy Function

We now investigate this entropy function of a distribution as a mathematical function independent of any meaning that may be assigned to the variables.

Example 7.3-1 The log Inequality

The first result we need is the inequality

$$\ln x \leq 1 - x \qquad (7.3\text{-}1)$$

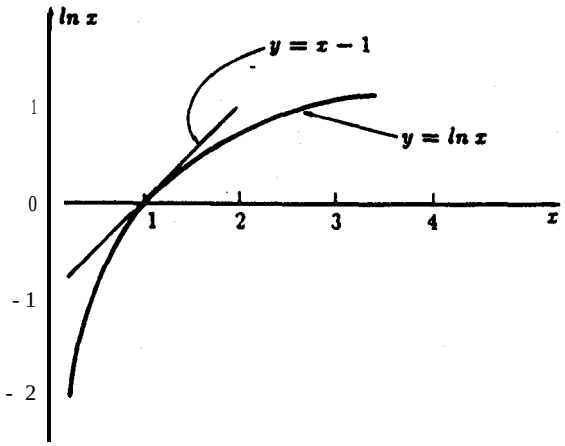


FIGURE 7.3-1

This is easily seen from Figure 7.3-1 where we fit the tangent line to the curve at $x = 1$. The derivative of $\ln x$ is $1/x$, and at $x = 1$ the slope is 1. The tangent line at the point (1,0) is

$$y - 0 = 1(x - 1)$$

or

$$y = x - 1$$

Since $y = \ln x$ is convex downward ($y'' = -1/x^2 < 0$) the inequaiity holds for all values of $x > 0$, and the equality holds *only* at $x = 1$. Compare with Appendix 1.B.

Example 7.3-2 ***Gibbs' Inequality***

We now consider two random variables X and Y with finite distributions $p_X(i)$ and $p_Y(i)$, both distributions with the sum of their values equal to 1. Starting with (using natural logs again)

$$\sum_{i=1}^{n} p_X(i) \ln \frac{p_Y(i)}{p_X(i)}$$

we apply the tog inequality (7.3-l) to each log term to get

$$\sum_{i=1}^{n} p_X(i) \ln \frac{p_Y(i)}{p_X(i)} \leq \sum_{i=1}^{n} p_X(i) \left\{ 1 - \frac{p_Y(i)}{p_X(i)} \right\} \tag{7.3-I}$$

$$\leq \sum_{i\,r\,l}^{n} p_X(i) - \sum_{i=1} y(i) = 1 - 1 = 0$$

The equality holds only when each $p_X(i) = p_Y(i)$. Since the right hand side is 0 we can multiply by any positive constant we want and thus, if we wish, convert to base 2. This is the important *Gibbs' inequality*.

$$\sum_{i=1}^{n} p_X(i) \log \frac{p_Y(i)}{p_X(i)} \leq 0 \tag{7.3-2}$$

The equality holds only when $p_X(i) = p_Y(i)$ for all *i.*

Example 7.3-3 The Entropy of Independent Random Variables

Suppose we have two independent random variables X and Y and ask for the entropy of their product. We have

$$H(XY) = \sum_{i}^{j} p_X(x_i) p_Y(y_j) \log \frac{1}{p_X(x_i) p_Y(y_j)}$$

$$= \sum_{i}^{j} p_X(x_i) p_Y(y_j) \log \frac{1}{p_X(x_i)} + \log \frac{1}{p_Y(y_j)}$$

When we sum over the variable not in the log term we get a factor of 1, and when we sum over the variable in the log term we get the corresponding entropy; hence

$$H(XY) = H(X) + H(Y) \tag{7.3-3}$$

When we compare this with (2.5-4)

$$E\{X + Y\} = E\{X\} + E\{Y\}$$



we see that the entropy of the product of independent random variables satisfies the same equation as the expectation on sums (independent or not), and partly explains the usefulness of the entropy function.

Suppose next that we have n independent random variables X_i, (i = 1, . . . , *n*), and consider the entropy of the compound event S = $X_1X_2 \ldots X_n$. We have, by similar reasoning,

$$H(S) = H(X_1) + H(X_2) + \cdots + H(X_n) \tag{7.3-4}$$

Thus n repeated trials of the same event X gives the entropy

$$H(S) = nH(X) \tag{7.3-5}$$

This is what one would expect when you examine the original definition of information carefully.

Example 7.34 *The Entropy Decreases When You Combine Items*

We want to prove that if we combine any two items of a distribution then the entropy decreases, that we lose information. Let the two items have probabilities $p(i)$ and p(j). Since probabilities are non-negative it follows that

$$p(i) \leq p(i) + p(j)$$

$$\log p(i) \leq \log\{p(i) + p(j)\}$$

$$\log\ 1/p(i) \geq \log \frac{1}{p(i) + p(j)}$$

Now: (1) multiply the last equation by p(i), (2) interchange i and j, and (3) add the two equations to get

$$p(i) \log \frac{1}{p(i)} + p(j) \log \frac{1}{p(j)} \geq \{p(i) + p(j)\} \log \frac{1}{p(i) + p(j)}$$

and we have our result which agrees with our understanding of Shannon's entropy: combining items loses information.

7.4

Exercises 7.3

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7.3-2 Wha
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$p_4 = 1/16$, p_5
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7.3-6 If yo
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7.3-7 Wha
7.3-8 Whe
7.3-9 Shov

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Example 7.4-1

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Exercises 7.3

7.3-1 What is the amount of information in drawing a card from a deck? Ans. log 52 = 5.7004397. . . bits/draw.

7.3-2 **What is the amount of information in drawing** k cards without re**placement from a deck?**

7.3-3 **Compute the entropy of the distribution** $p_1 = 1/2$, $p_2 = 1/4$, $p_3 = 1/8$, $p_4 = 1/16$, $p_5 = 1/16$.

7.3-4 **Write a symmetric form of Gibbs' inequality.**

7.3-5 Show that when combining two items having the same probability p(i) you lose $-2p(i) \ln 2$ **of information.**

7.3-6 If you have a uniform distribution of 2n items and combine adjacent **terms thus halving the number of items, what is the information loss?**

7.3-7 **What is the information in the toss of a coin and the roll of a die?**

7.3-8 What is the information from the roll of n independent dice?

7.3-9 Show that the information from the toss of n independent coins = n.

7.4 Some Simple Applications

We now show that Shannon's definition of entropy has a number of other reasonable consequences that justify our special attention to this function.

Example 7.4-1 ***Maximum Entropy Distributions***

We want to show that the distribution with the maximum entropy is the flat distribution. We start with Gibbs' inequality, and assign the $p_Y(i) = 1/n$ (where n is the number of items in the distribution). We have

$$\sum_i p_X(i) \log \frac{1}{n\, p_X(i)} \leq 0$$

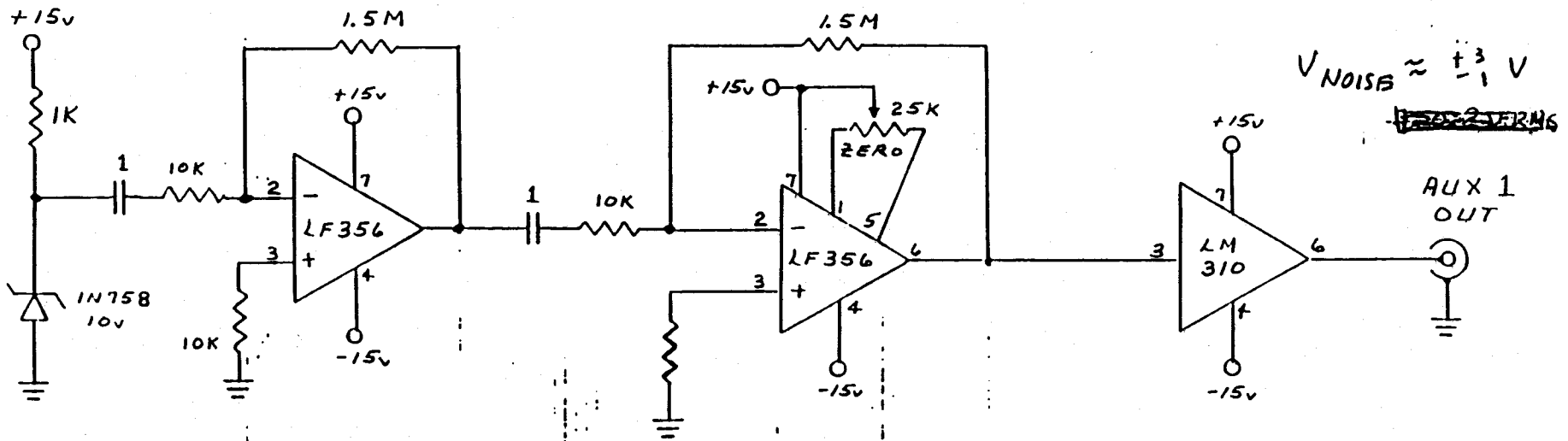
Write the log term of the product as the sum of logs and transpose the terms in n. We have

$$H(X) = \sum_i p_X(i) \log \frac{1}{p_X(i)} \leq \sum_i p_X(i) \log n = \log n$$

or

$$H(X) \leq \log n$$

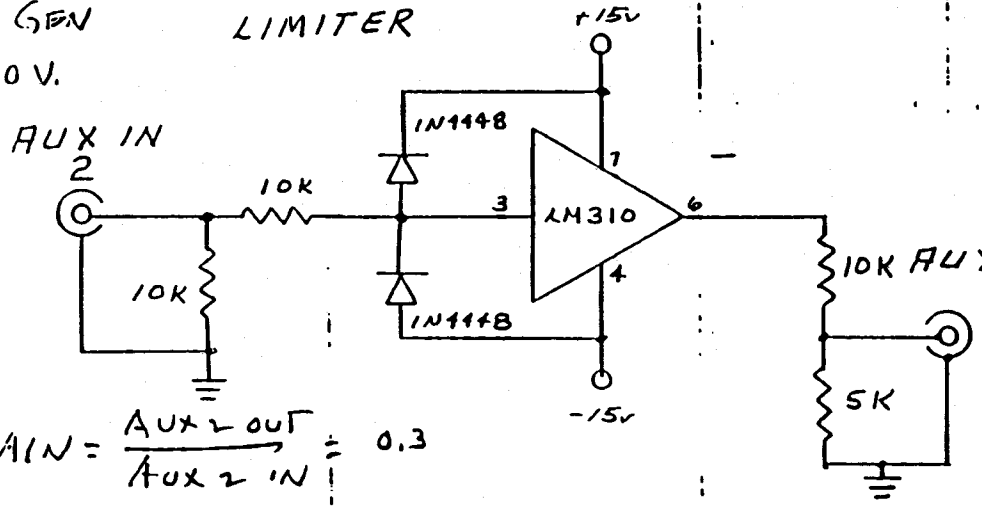
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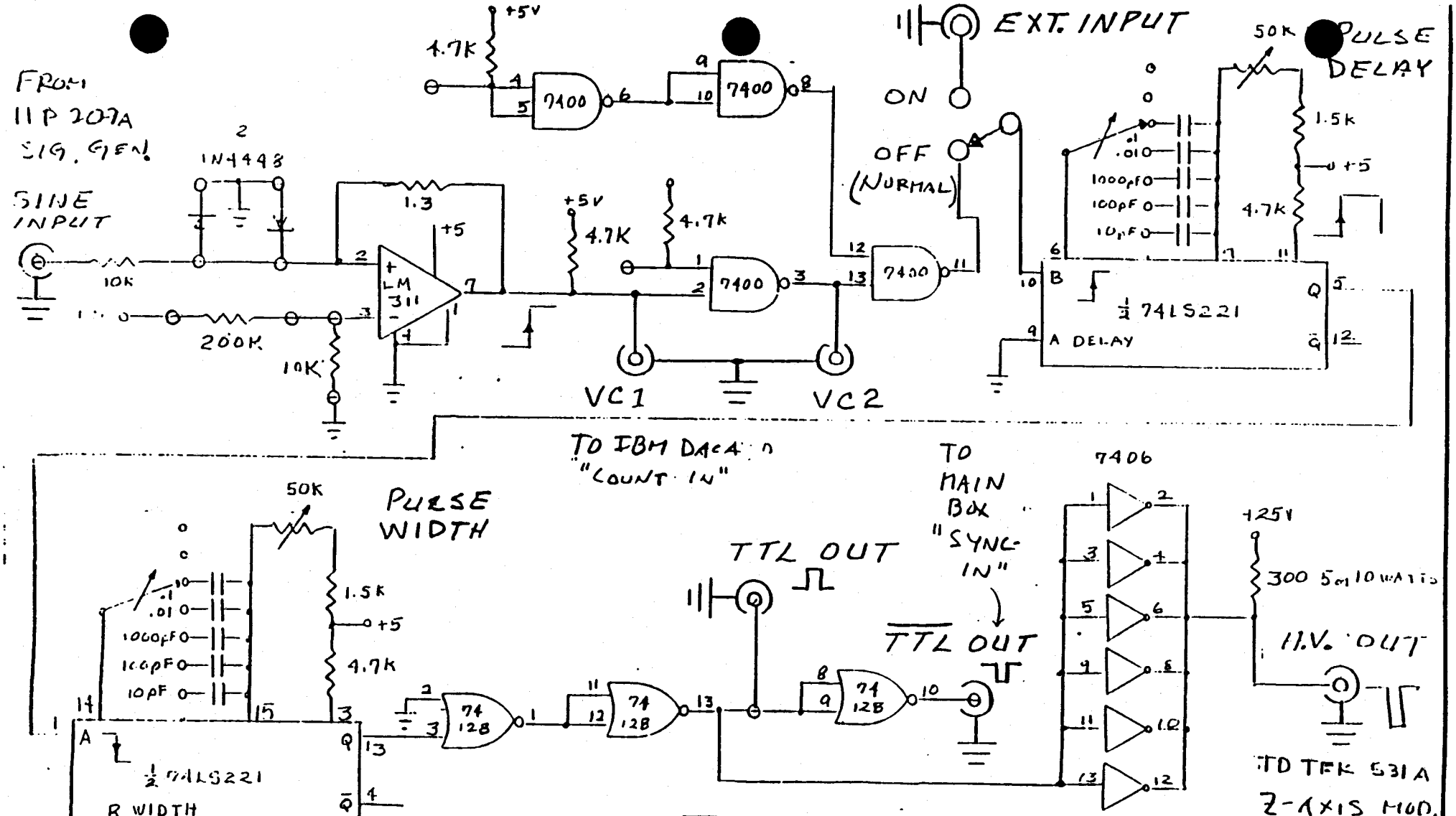
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Nonlinear Dynamics Experiment

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4.3.2



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.1μF	= 3mS
1μF	= 30mS
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NONLINEAR DYNAMICS EXPERIMENT

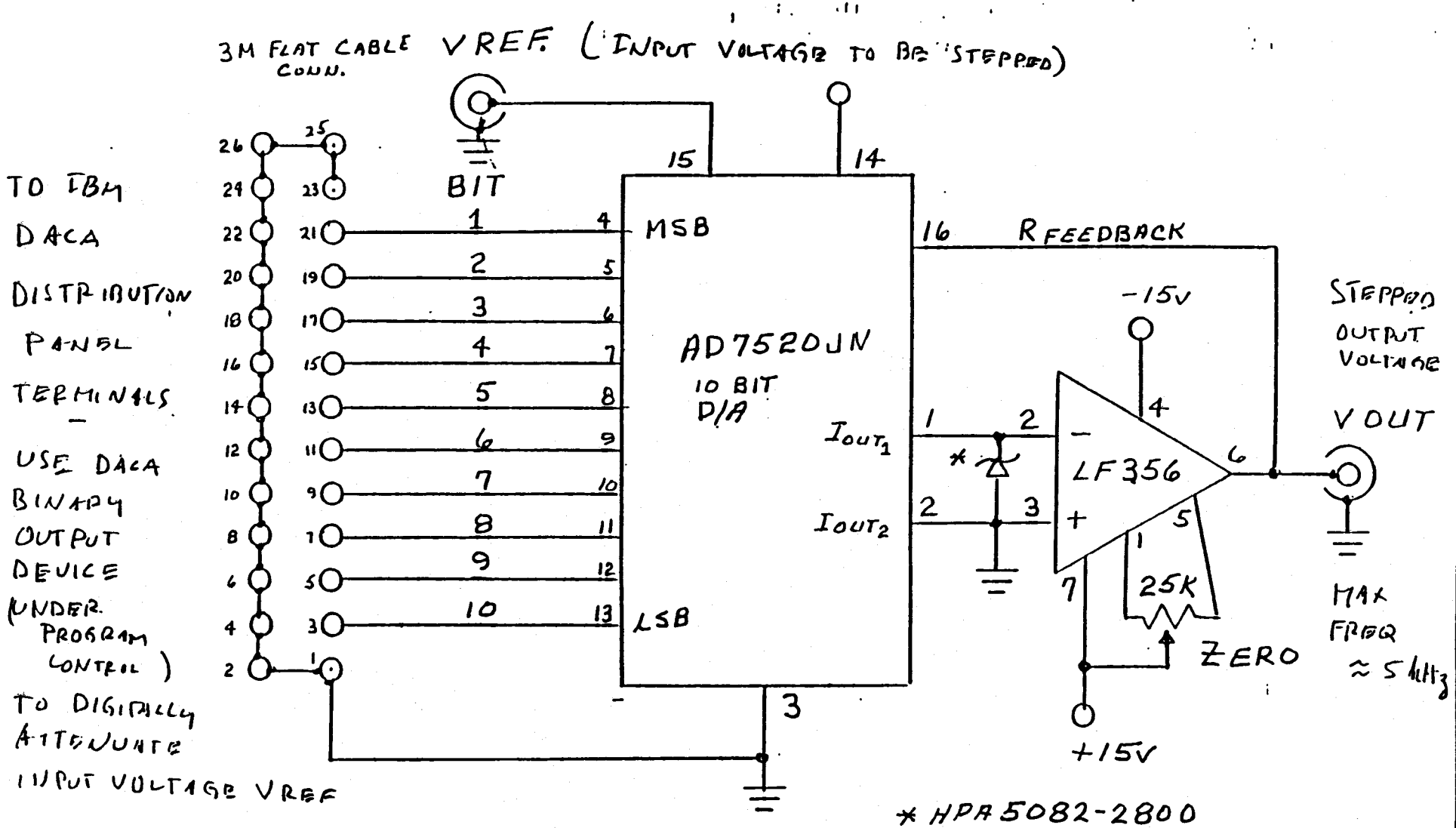
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SCALE:	APPROVED BY: C. D. Jeffries	DRAWN BY
DATE:		REVISED

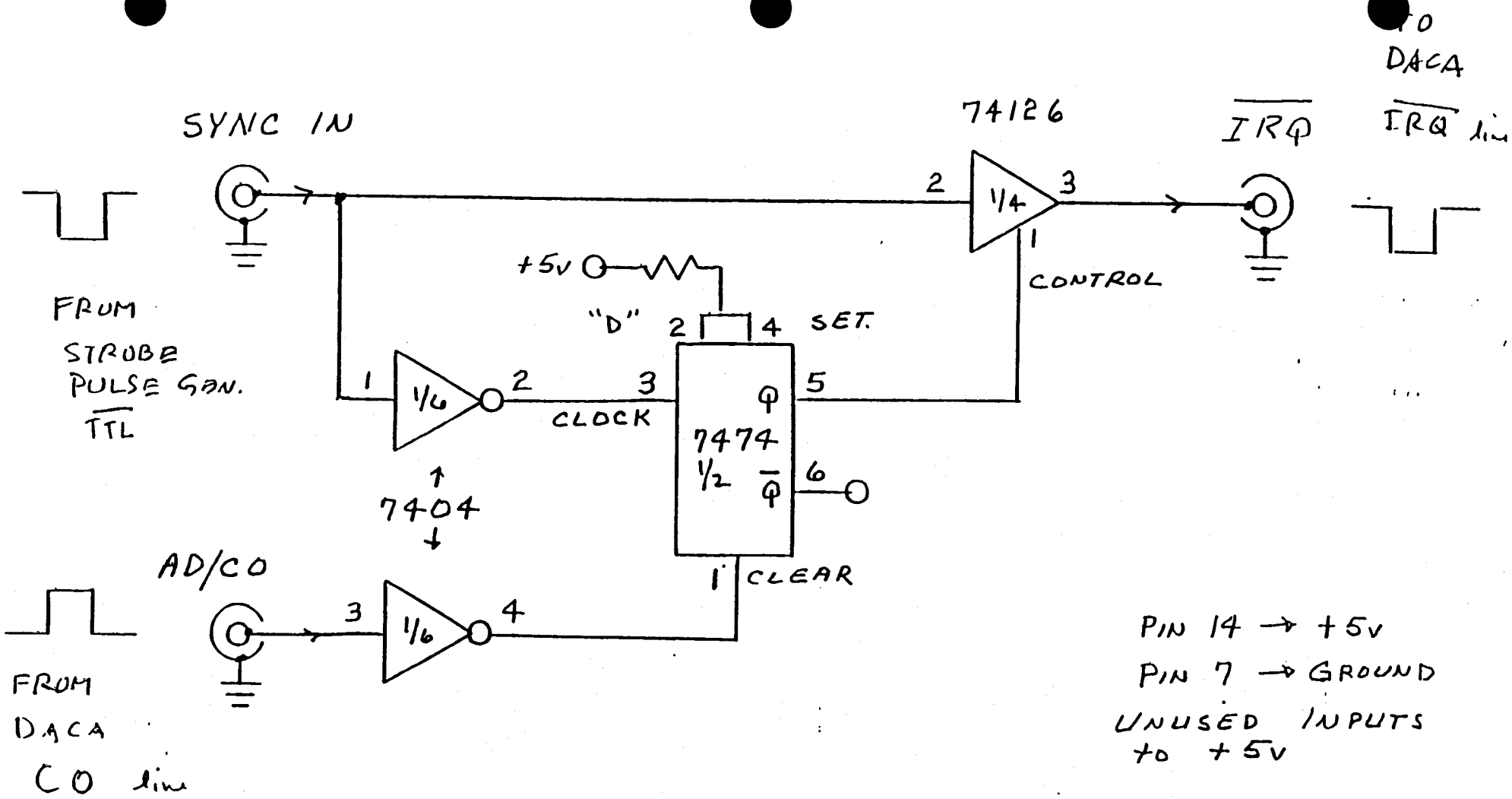
University of California Physics Dept.

Electronics Shop 4.3.4

DRAWING NUMBER 836-35

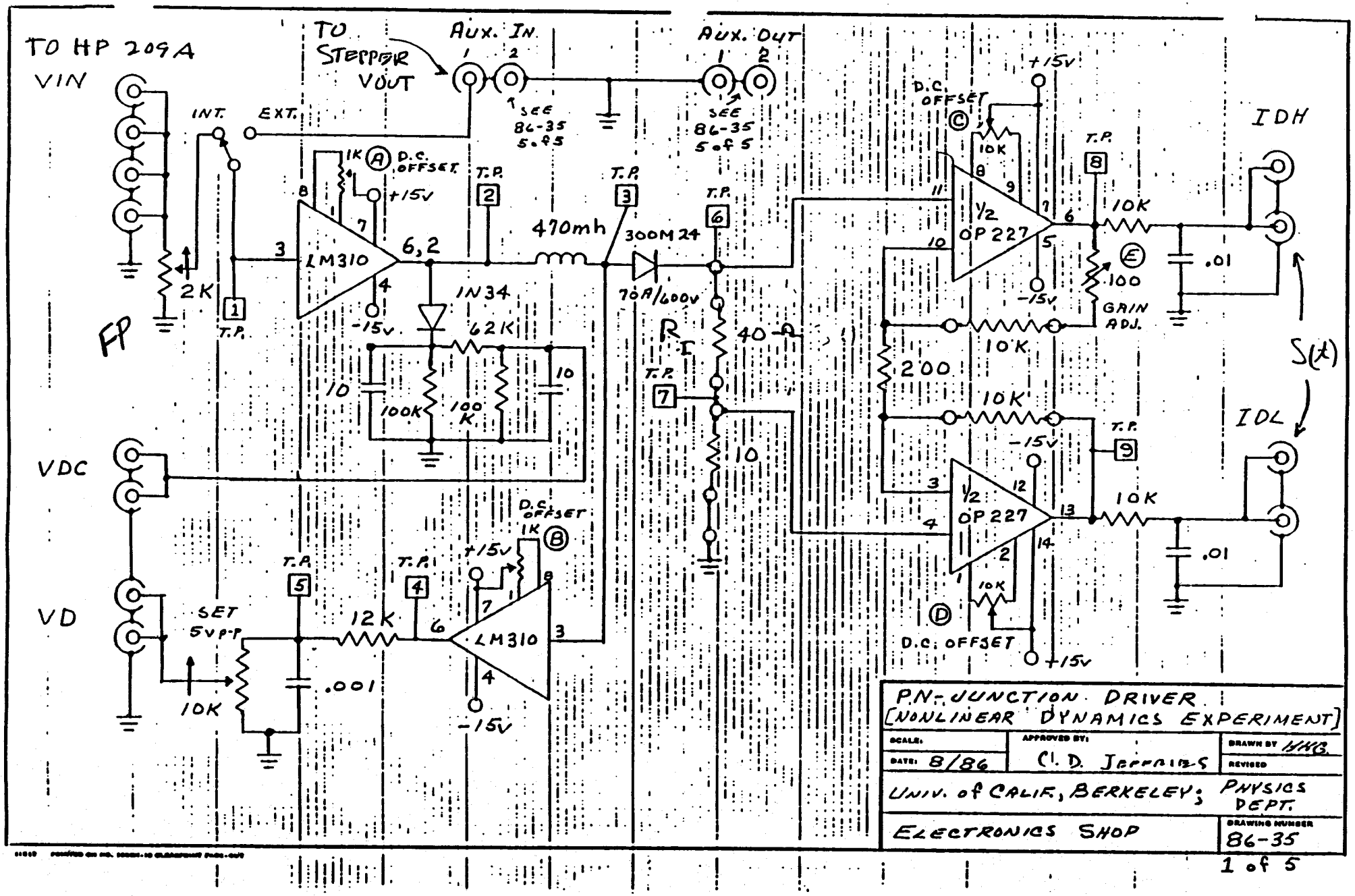


STEPPER CARD		
NONLINEAR DYNAMICS EXPERIMENT		
SCALE:	APPROVED BY: C D Jeffries	DRAWN BY 15/16B
DATE: 8/86		REVISED
UNIV. OF CALIF., BERKELEY; PHYSICS DEPT.		
ELECTRONICS SHOP		DRAWING NUMBER 8125

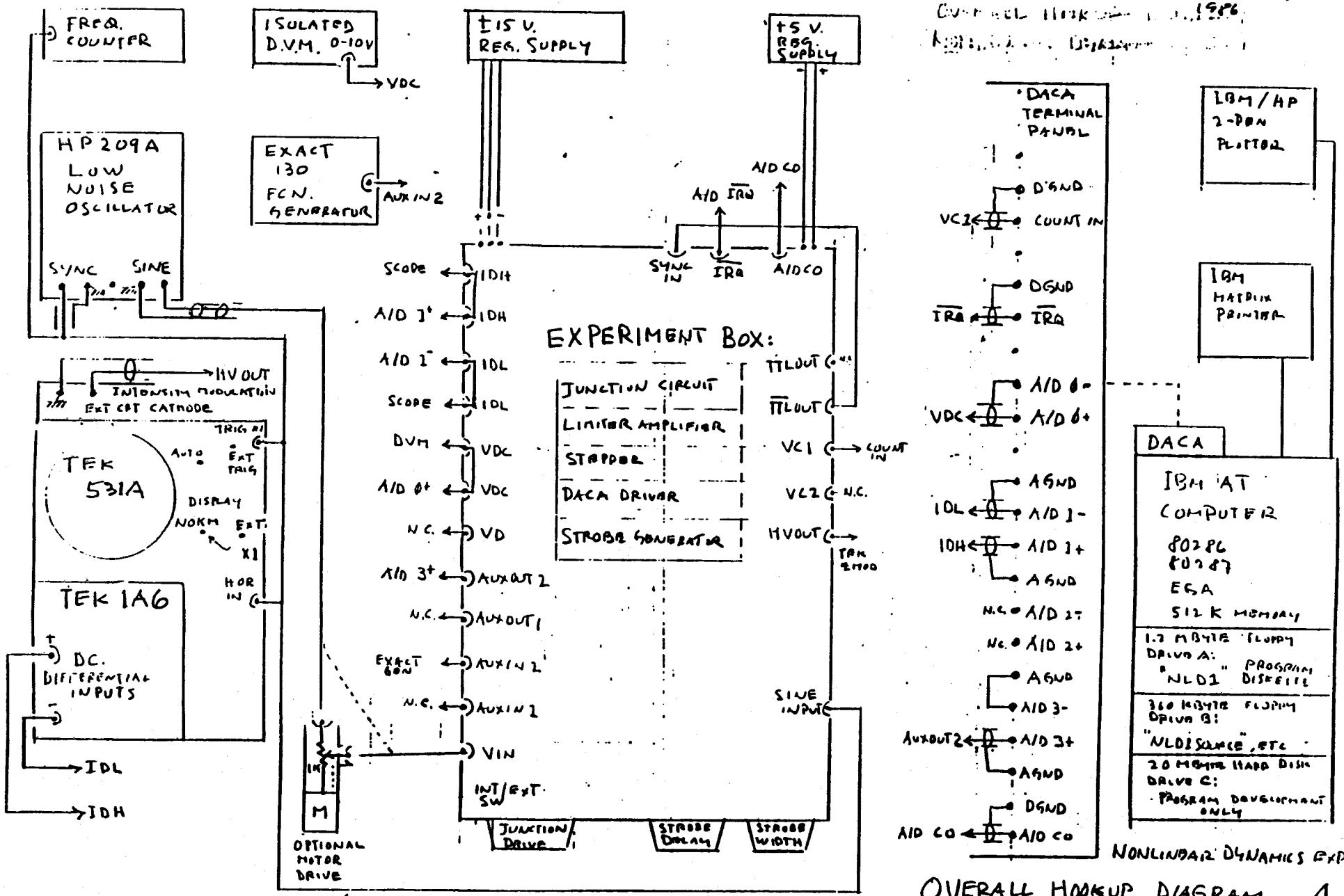


IBM DACA TRIG. CARD NONLINEAR DYNAMICS EXPERIMENT		
SCALE:	APPROVED BY: Cb Jaffrian	DRAWN BY KHG
DATE: 8/86		REVISED
UNIV. of CALIF., BERKELEY; PHYSICS DEPT.		
ELECTRONICS SHOP		DRAWING NUMBER 86-35

11"x17" PRINTED ON NO. 1000H-10 CLEARPRINT FADE-OUT

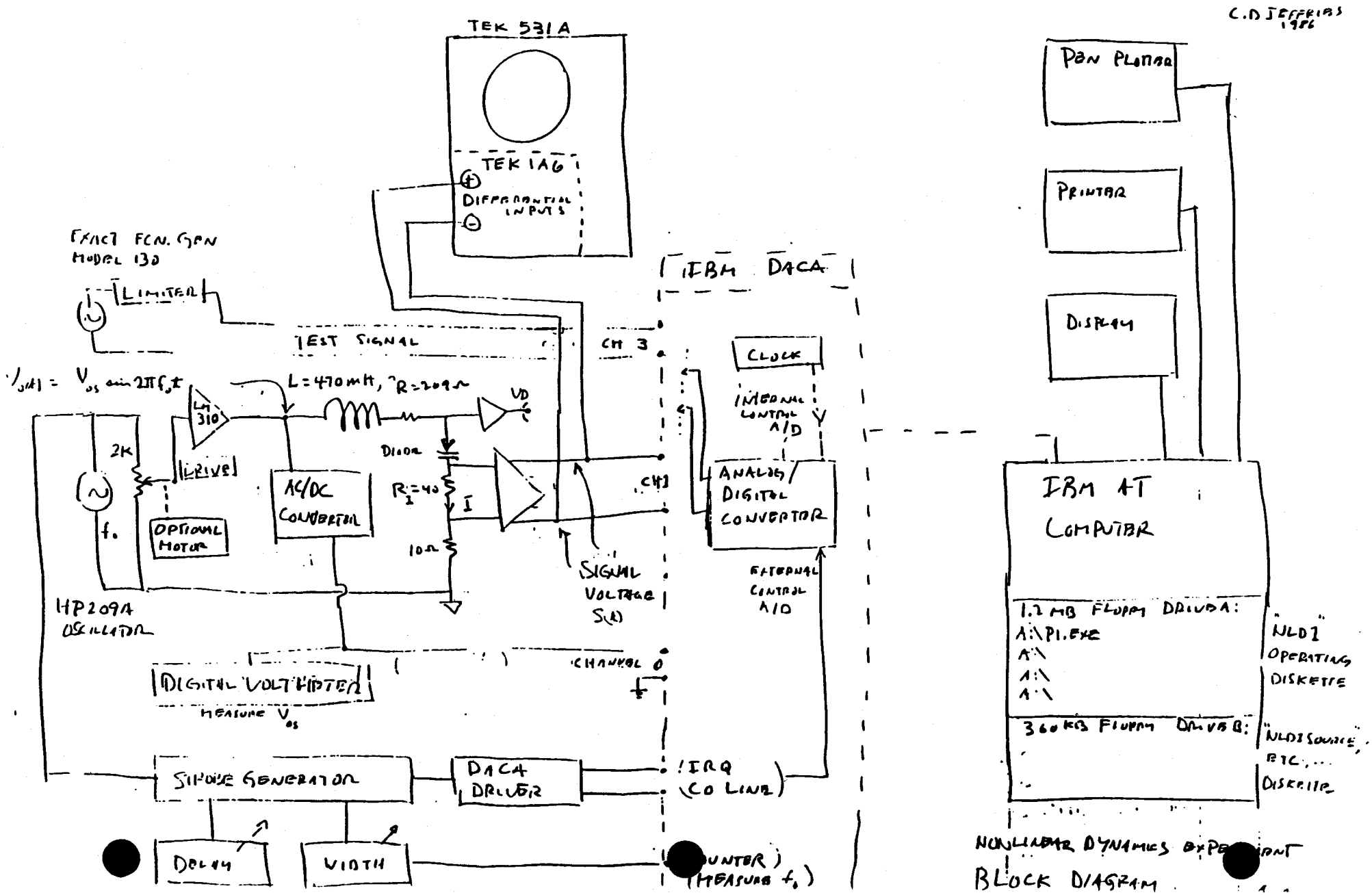

TO HP 209A
VIN
INT.
EXT.
2K
T.P. 1
FP
TO STEPPER
VOUT
AUX. IN
1
2
SEE 86-35 5 of 5
AUX. OUT
1
2
SEE 86-35 5 of 5
1K (A) D.C. OFFSET
+15V
T.P. 2
T.P. 3
T.P. 6
470mh
300M24
70A/600V
8
1
7
3
LM310
6,2
4
-15V
IN34
62K
10
100K
100K
10
R
T.P. 7
40
10
VDC
VD
SET 5Vp-p
10K
T.P. 5
T.P. 4
12K
.001
1K (B) D.C. OFFSET
+15V
LM310
-15V
+15V
D.C. OFFSET (C)
10K
T.P. 8
11
8
9
1/2 OP 227
10
7
6
5
-15V
10K
100
GAIN ADJ.
.01
IDH
S(t)
200
10K
-15V
T.P. 9
3
12
1/2 OP 227
4
13
14
2
1
10K
(D) D.C. OFFSET
+15V
10K
.01
IDL
P.N. JUNCTION DRIVER
[NONLINEAR DYNAMICS EXPERIMENT]
SCALE:
APPROVED BY:
C. D. JEFFRIES
DRAWN BY WMG
DATE: 8/86
REVISED
UNIV. of CALIF, BERKELEY; PHYSICS DEPT.
ELECTRONICS SHOP
DRAWING NUMBER 86-35
1 of 5

4.3.1



CARSON-JOFFRIES

OVERALL HOOKUP DIAGRAM 4.2


C.D JEFFRIES 1986
TEK 531A
TEK 1A6
DIFFERENTIAL INPUTS
EXACT FCN. GEN MODEL 130
LIMITER
TEST SIGNAL
CH 3
IBM DACA
CLOCK
INTERNAL CONTROL A/D
$V_{sig} = V_{os} \sin 2\pi f_0 t$
L = 470mH, R = 209Ω
LM 310
VD
2K
DRIVE
DIODE
AC/DC CONVERTER
R_1 = 40
I
CH1
ANALOG/DIGITAL CONVERTOR
OPTIONAL MOTOR
f_0
10Ω
SIGNAL VOLTAGE S(t)
EXTERNAL CONTROL A/D
HP2094 OSCILLATOR
CHANNEL 0
DIGITAL VOLTMETER
MEASURE V_{os}
SINUSE GENERATOR
DAC4 DRIVER
!IRQ (CO LINE)
DELAY
WIDTH
COUNTER (MEASURE f_0)
PEN PLOTTER
PRINTER
DISPLAY
IBM AT COMPUTER
1.2 MB FLOPPY DRIVE A:
A:\P1.EXE
A:\
A:\
A:\
"NLD1 OPERATING DISKETTE"
360 KB FLOPPY DRIVE B:
"NLD1 SOURCE, ETC. ... DISKETTE"
NONLINEAR DYNAMICS EXPERIMENT
BLOCK DIAGRAM